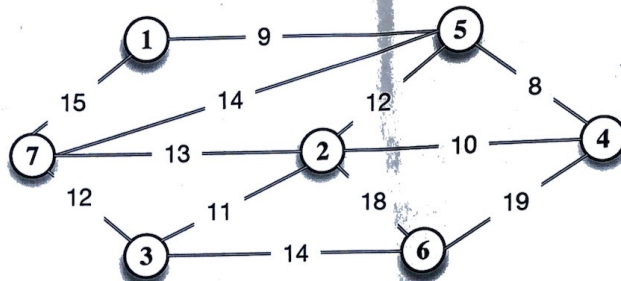


First part : Closed book

Total score: 17 pts. Exercise scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

One of the exercises indicated with * can be skipped. The score will be provided by the minimum between the exercise score and the bonus points gained during the course. Please indicate which of the exercises you want to skip

1. * Solve the Minimum Spanning Tree for the graph in the figure, with the costs associated to the arcs. Specify the algorithm that you apply and its complexity. Show the intermediate steps.



Consider a parametric costo of arc (2,3) equal to $11+\alpha$. Determine the minimum value of α that makes (2,3) exit the optimal spanning tree? Why? What is new tree?

2. Consider the following linear programming problem:

$$\begin{aligned} \max \quad & 6x_1 + 21x_2 \\ & -4x_1 + 6x_2 \leq 15 \\ & x_1 + 3x_2 \leq 12 \\ & 2x_1 \leq 9 \\ & -2x_1 - 2x_2 \leq -3 \end{aligned}$$

Write the dual and the complementary slackness equations. For each of the following solutions $x' = \begin{bmatrix} 9/2 \\ 11/2 \end{bmatrix}$,

$x'' = \begin{bmatrix} 3/2 \\ 7/2 \end{bmatrix}$ derive the complementary dual solutions and discuss their primal and dual feasibility.

3. * Consider the following Integer Linear Programming problem:

$$\begin{aligned} \min \quad & x_1 + 2x_2 \\ & -2x_1 - 2x_2 \leq -14 \\ & 2x_1 - 2x_2 \leq -3 \\ & x_1, x_2 \in \mathbb{Z}_+ \end{aligned}$$

Add the slack variables (x_3 and x_4) to the first and the second constraints, and consider the optimal basis of the LP relaxation $B = \{1, 2\}$. Generate the Gomory cuts associated with the fractional components of the solution. Justify your answer and give the geometric representation of the feasible region and of the cut(s).

Consider the cut associated with x_2 . Is there a way to strengthen it maintaining the validity of the inequality? Justify your answer.

4. * Consider the LP problem $P : \min cx : Ax \leq b, x \geq 0, x \in \mathbb{R}^n$. Write the dual D using y to denote the variable vector. If needed, use the conversion table in the back for writing the dual. Given two feasible solutions $\bar{x}$ and $\bar{y}$ for P and D , respectively, say which of the following statements is always true and justify the answer: i) $c\bar{x} = \bar{y}b$, ii) $c\bar{x} \leq \bar{y}b$, iii) $c\bar{x} \geq \bar{y}b$

Second part: Open book

Total score: 10 pts. In case you made the “big” lab project this part can be skipped. If you return this part, the “big” project points will be cancelled.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. A multinational automobile company can to produce and market n different car models. The CEO must decide which models to produce and bring to market in the next year. The multinational corporation has d divisions, each of which is involved to varying degrees in activities related to each car model. The coefficient a_{ij} gives the number of managers of division j needed to activate the production and marketing of model i . Each division j has b_j managers available, and no more are expected to be hired. Each model i has a market penetration equal to p_i . The CEO must decide which models to produce while taking into account available resources and maximizing market penetration. Furthermore, let S be the subset of models in the SUV class, for image reasons, the CEO does not want to produce more than s models in this class. Formulate the CEO's decision problem in terms of integer linear programming. Specify variables, constraints and objective function explaining their meaning.

Variant: let H be the subset of hybrid models. Each hybrid model i has a corresponding full electric model i' . If both are produced there is a resource saving of r_{ij} for each division j . How does the formulation change? Be careful to maintain linearity in the formulation.

6. During the COVID emergency, a city must set up centers to perform PCR testing. The city has n neighborhoods and m health centers in which to perform testing. The residents of each neighborhood i can reach a subset of C_i centers. The estimated number of daily tests in each neighborhood i is t_i . Each center j can perform at most u_j tests per day. Formulate the problem of whether the capacity to perform the required tests is sufficient in terms of maximum flow on an appropriate graph. Specify nodes, arcs, and capacity of the arcs. Considering the optimal solution of the maximum flow, in which case is the answer affirmative?

Variant: Suppose that the capacity of the system is not sufficient. The city health manager must then resort to private pharmacies to meet the excess demands. Each center j can call on pharmacies in the area to do tests at a cost of c_j per test. Formulate the variant of the problem as a minimum cost flow. How is the graph modified? Remember to specify the balances of the nodes.